STATISTICAL MODELLING AND DESIGN OF EXPERIMENTS

Assignment 1

Joel González Jiménez

Jose Arias Perez

Quentin Meneghini

14th April 2026

Facultat d'Informàtica de Barcelona

Contents

Introduction **1**

1 Questions 1 **2**

1.1 Q1-a 2

1.2 Q1-b 3

1.3 Q1-c 4

1.4 Q1-d 6

1.5 Q1-e 6

1.6 Q1-f 6

1.7 Q1-g 7

1.8 Q1-h 8

1.9 Q1-i 8

1.10 Q1-j 9

1.11 Q1-k 9

2 Questions 2 **11**

2.1 Q2-a 11

2.2 Q2-b 12

2.3 Q2-c 14

2.4 Q2-d 14

2.5 Q2-e 15

Conclusion **17**

A Appendix Chapter **18**

Introduction

Chapter 1

Questions 1

1.1 Import data set to R assigning the type of each variable correctly

We casted to factor both the string types [”Model”, ”SalesRegion”] and the very repetitive columns [”NumberofEngines”, ”EngineType”]. We considered casting also ”ProductionYear”, but in later calculation as the median was required, so we decided to leave as is (numerical). The code is shown below.

```
air_data <- read.csv("./dataset.csv", sep=",", stringsAsFactors=TRUE)

air_data <- air_data %>%
  rename(
    FC = FuelConsumption.L.h.,
    HM = HourlyMaintenance...,
    Price = Price...
  )

to_factors <- c("Model", "NumberofEngines", "EngineType", "SalesRegion")

for (col in to_factors) {
  air_data[[col]] <- as.factor(air_data[[col]])
```

```
}
```

1.2 Summarize the variables Price, Fuel Consumption first for the all data and then for the groups of Engine Type

We reported the result of the summaries of both Price and Fuel Consuption in the tables 1.1 and 1.2, while an histogram is in figure 1.1

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
145815	14096814	83921914	198833650	384323881	978213229

Table 1.1: Price

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2.00	5.95	9.82	12.08	13.47	49.97

Table 1.2: Fuel Consumption (FC)

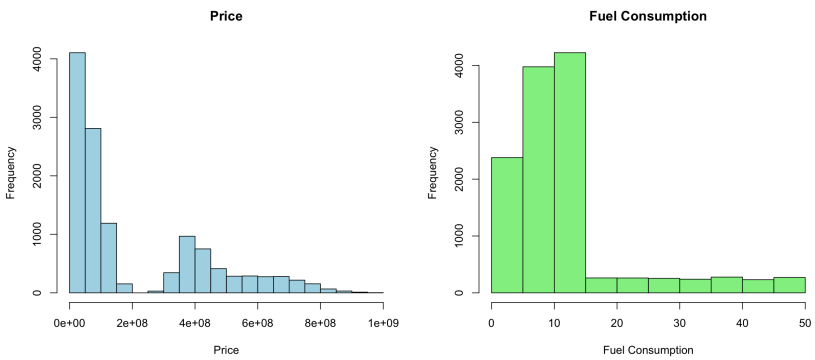

Figure 1.1: Histograms

Here are the same summaries but based on the Engine Type (Table 1.3 and Figure 1.2.

EngineType	Price_Mean	Price_SD	Price_Median	Price_Min	Price_Max	FC_Mean	FC_SD	FC_Median	FC_Min	FC_Max	n
Piston	279405.	77369.	251474.	145815.	561279.	30.1	11.6	29.9	10.0	50.0	2039
Turbofan	237995200.	231292861.	108876395.	8150222.	978213229.	8.52	3.75	8.53	2.0	15.0	10338

Table 1.3: Summary by Model

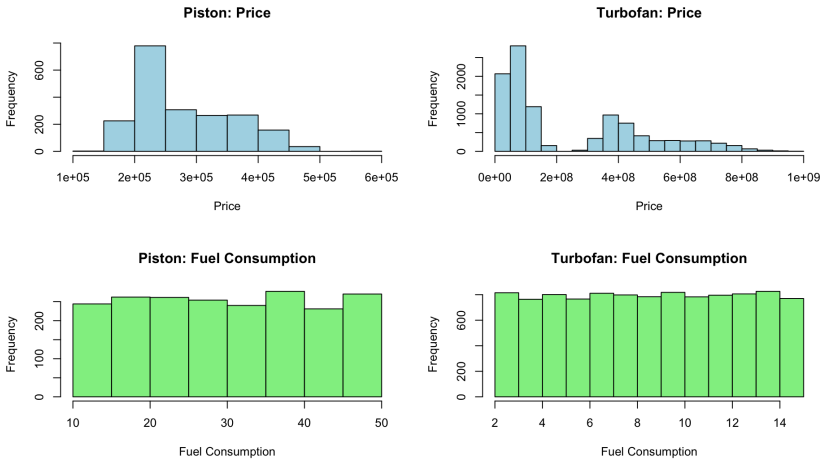

Figure 1.2: Summary distributions by Model

1.3 Test whether Fuel Consumption is affected from the Engine Type of the plane. Check the assumptions and visualize the relationship between these two characteristics

```
        Welch Two Sample t-test

data:  FC by EngineType
t = 83.347, df = 2123.1, p-value < 2.2e-16
alternative hypothesis: true difference in means between group Piston and group Turbofan is not equal to 0
95 percent confidence interval:
 21.09784 22.11459
sample estimates:
  mean in group Piston mean in group Turbofan
             30.122398               8.516185
```

Figure 1.3: T-test results for Fuel Consumption by Engine Type

We performed a t-test and obtained a p-value, the results can be found in the figure 1.3. Since the p-value is < 0.05, we reject the null hypothesis and conclude that there is a significant difference in mean fuel consumption between the engine types. Furthermore we used a visual interpretation to be sure about the result as the number of observation made the test not very significant (figures 1.4 and 1.5).

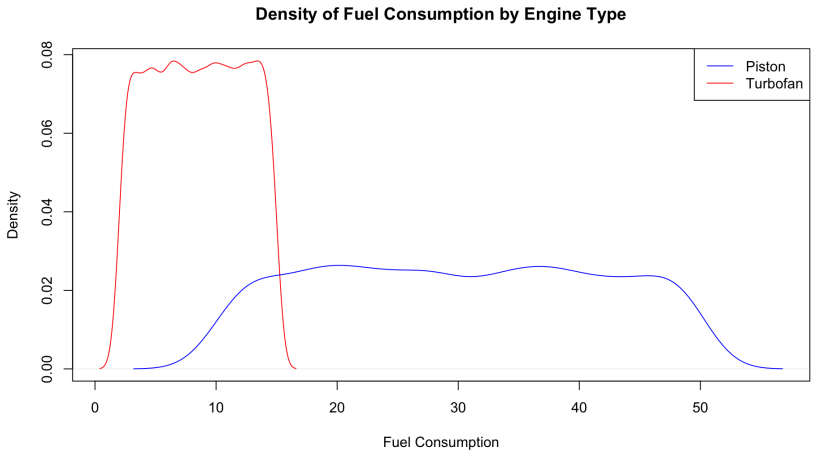

Figure 1.4: Comparing distributions

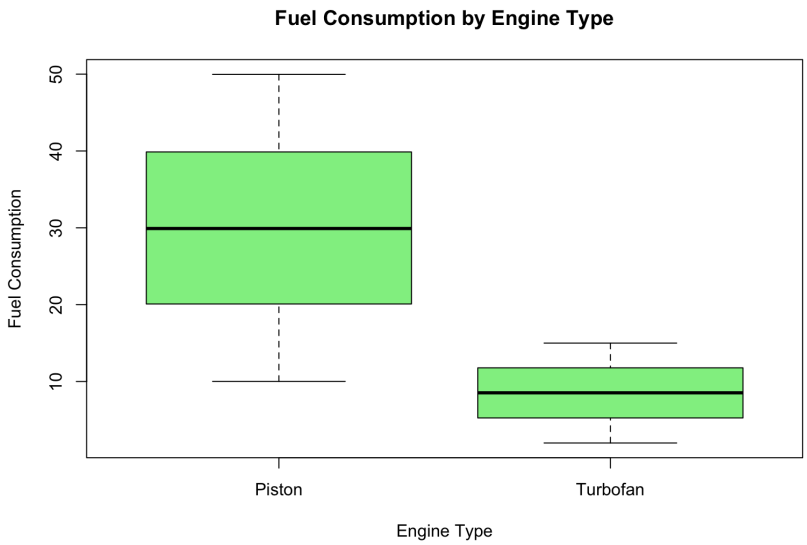

Figure 1.5: Boxplot is best for comparing a continuous variable across categories

Test assumptions

- We checked for normality (Shapiro-Wilk) and homogeneity of variances (Levene's test). Because the sample size is large (N > 5000), the normality test is overly sensitive, but the large sample size allows us to rely on the Central Limit Theorem.
- Variance was addressed by using the default robust t-test in R (Welch's t-test).

1.4 Construct 95% confidence intervals for the mean of two groups and interpret them

- We are 95% confident that the true population mean of FC for Piston engines lies within that interval **(29.61919, 30.62560)**.
- Similarly, the 95% CI for Turbofan is **(8.443816, 8.588554)**.

The two intervals do NOT overlap. This provides visual evidence that the mean Fuel Consumption differs significantly between the two engine types.

1.5 Check the association between Model and Sales Region in the whole sample using proper method and interpret your findings

To check the association between the 2 categories we used the chi square as the two types of variable needed it. The code is shown below.

```
table_model_region <- table(air_data$Model, air_data$SalesRegion)
chisq_result <- chisq.test(table_model_region)
```

The obtained p-value > 0.05 ($\sim$0.73), so we fail to reject the null hypothesis and conclude that there is no statistically significant association between Model and Sales Region. The airplane models are distributed similarly across sales regions.

1.6 Filter your data only considering $BombardierCRJ200$ and $Cessna172$ model airplanes

Here we applied a simple filter and removed the empty buckets. this way we obtained a clean dataset with only the 2 models.

```
filtered_air_data <- air_data |>
  filter(Model %in% c("$Bombardier CRJ200$", "$Cessna 172$")) |>
  droplevels()
```

1.7 Check the distribution of Price across two categories of Engine type (You can apply transformation if you think it is required)

Here we applied a transformation to the price (log) since it appeared to be completely left-skewed. visually we can clearly conclude that there is a significant difference of Price between the two types of Engines. Data can be visualized in figures 1.6 and 1.7

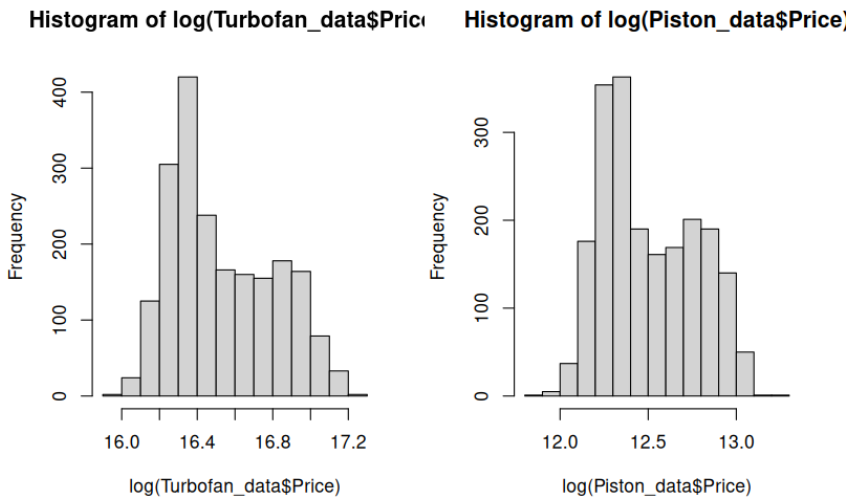

Figure 1.6: Histogram of Log Price by Engine Type

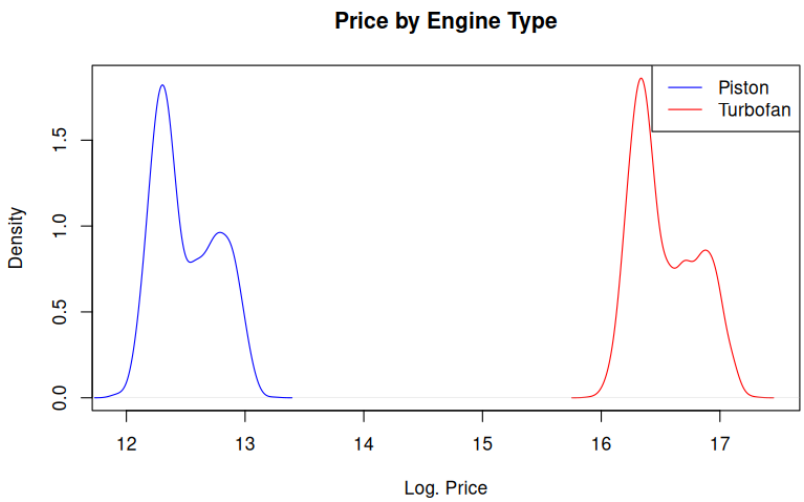

Figure 1.7: Distribution plot of Log Price by Engine Type

1.8 Categorize the variable Price into two categories as "low" and "High" by cutting from the median and save it as a new variable into your data frame

To compute the categorization of price we first calculated the median and then applied it to a new variable: *Price_Category*.

```
median_price <- median(filtered_air_data$Price, na.rm = TRUE)
filtered_air_data$Price_Category <- factor(
  ifelse(filtered_air_data$Price > median_price, "High", "Low")
)
```

1.9 Cross classify model and price categories and interpret the conditional probabilities

Row proportions P(PriceCategory | Model) table shows (table 1.4), for each model, what proportion of its airplanes fall into "Low" vs "High" price. Since $BombardierCRJ200$ (Turbofan) has much higher prices than $Cessna172$ (Piston), we expect nearly all $BombardierCRJ200$ to be in the "High" category and nearly all $Cessna172$ to be in the "Low" category. Nevertheless, using the median as a separator, some $BombardierCRJ200$ still fell into the "low" category because the number of the first model is slightly higher than that of the second model..

	High	Low
$BombardierCRJ200$	0.997074598	0.002925402
$Cessna172$	0.000000000	1.000000000

Table 1.4: Conditional probabilities: P(PriceCategory | Model) - row proportions

Column proportions P(Model | PriceCategory) table shows (table 1.5), for each price category, what proportion belongs to each model. This table confirms further the hypothesis of association between the model and price. The "low" category is dominated by $Cessna172$ and the "High" category by $BombardierCRJ200$, so it confirms a strong association between airplane model and price

	High	Low
$BombardierCRJ200$	1.000000000	0.002933985
$Cessna172$	0.000000000	0.997066015

Table 1.5: Conditional probabilities: P(Model | PriceCategory) - column proportions

1.10 Is there an association between the model of the airplane and its price level. Analyze it by using proper statistical method

we applied the chi square test and obtained a p-value as a result.

```
test_model_price <- chisq.test(cross_model_price)
```

The p-value < 0.05, so we reject the null hypothesis and conclude that there is a statistically significant association between airplane model and price level. Given that $BombardierCRJ200$ (Turbofan) costs millions while $Cessna172$ (Piston) costs hundreds of thousands, we expect a very small p-value, confirming a strong association.

1.11 Cross classify the variables Model and Sales region. Interpret the conditional probabilities and then test whether there is an association between these two characteristics

P(SalesRegion | Model) (row proportions, table 1.6) shows the distribution of sales regions for each model. Both models have similar proportions across regions, so we conclude that the sales region is independent of the model.

	Africa	Asia	Australia	Europe	North America	South America
$BombardierCRJ200$	0.1843003	0.1628474	0.1604096	0.1589469	0.1638225	0.1696733
$Cessna172$	0.1652771	0.1726336	0.1696910	0.1765571	0.1574301	0.1584110

Table 1.6: Conditional probabilities: P(SalesRegion | Model)

P(Model | SalesRegion) (column proportions, table 1.7) shows, within each region, what share belongs to each model. Both models appear in roughly equal proportions within every region, so this supports our independence assumption.

	Africa	Asia	Australia	Europe	North America	South America
$BombardierCRJ200$	0.5286713	0.4868805	0.4874074	0.4752187	0.5114155	0.5186289
$Cessna172$	0.4713287	0.5131195	0.5125926	0.5247813	0.4885845	0.4813711

Table 1.7: Conditional probabilities: P(Model | SalesRegion)

We applied a chi square test to further test the hypothesis. As expected, p-value > 0.05 ($\sim$0.29), so we fail to reject the null hypothesis and conclude there is no significant association. The two models are sold in similar proportions across regions.

Chapter 2

Questions 2

2.1 Create a new data frame only including "Airbus $A320$", "Airbus $A350$", "Boeing 737" and "Boeing 777" models of airplanes. Check the distribution of Price: first for the observations in this sample and then for each model in the data frame. Interpret your findings.

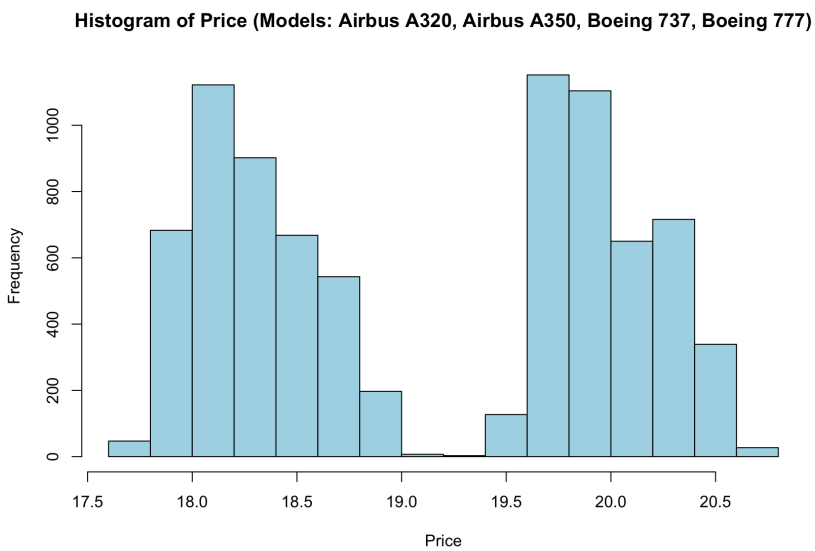

Figure 2.1: Price for all selected models

After the log transformation, Price is approximately normally distributed overall. Per model, Boeing 777 and Airbus $A350$ have higher mean log(Price) values reflecting their larger capacity

and longer range. Boeing 737 and Airbus $A320$ cluster at lower price levels. Within each model, the log(Price) distribution is roughly symmetric, confirming that log transformation was appropriate for subsequent ANOVA analysis.

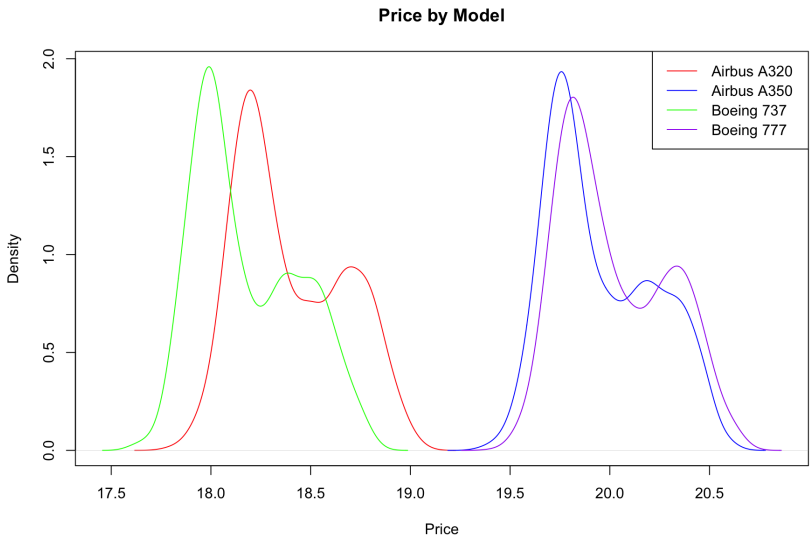

Figure 2.2: Price by model

2.2 Analyze the numerical variables that are affected by the "Model". Test the assumptions of the statistical method, for the cases that you have found a significant association, by using corresponding tests and plots. Write your conclusions.

We first used a visual approach to filter the variables that had no significance. The boxplots in figure 2.3 show that the four models fall into two sub-groups: cheaper and more expensive models. This difference created a change in some variables, but left untouched others. The next section highlights our analysis.

Variables Affected by Model

- **Price**: The $A350$ and $B777$ (expensive sub-group) are in a different wealth bracket compared to the $A320$ and $B737$ (cheaper sub-group).
- **Capacity and RangeKm**: These are affected by the model, though there is almost zero

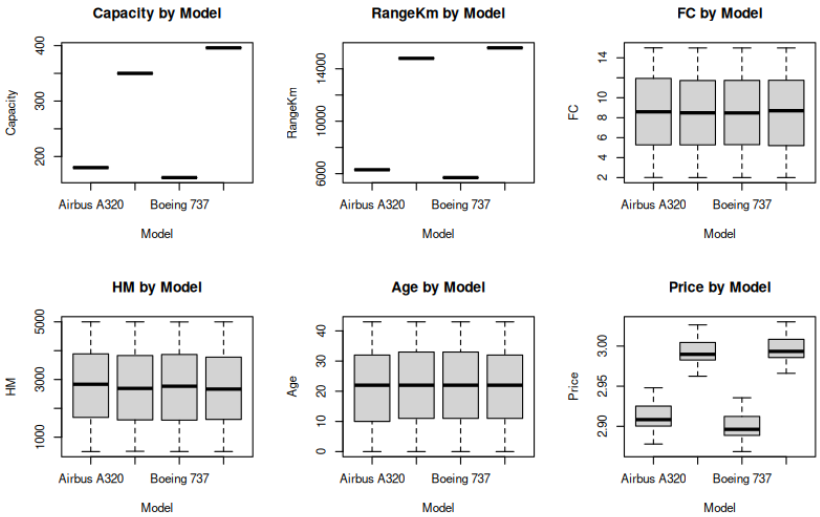

Figure 2.3: Boxplot of numerical variables

variation within groups in the dataset. This means Model, Capacity, and RangeKm are all telling the model the same thing (high multicollinearity).

- **FC, HM, and Age**: NOT significantly different across models since all four are Turbofan jets and the dataset appears to have similar distributions for these.

Assumption Analysis

We analyzed only the variable that passed the visula test.

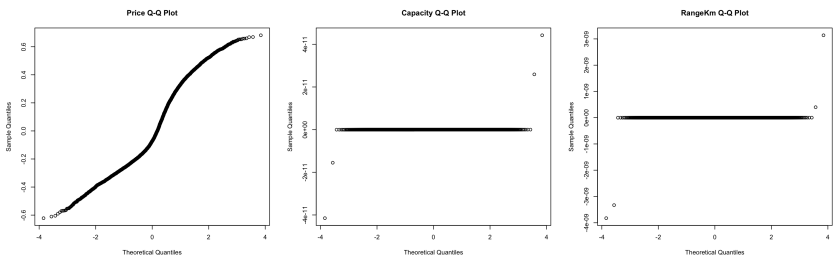

Figure 2.4: QQ-plot for all numerical variables

- The appearance of a "horizontal flat line" in the QQ-plots for **Capacity** and **RangeKm** occurs because these are fixed specifications for a given airplane model. Consequently, there is near-zero variance within each model group.

- For the case of **Price** the QQ-plot looks a bit curved with respect to the line meaning it might not be normally distributed. However, both Independence test and Homoscedasticity test show a p-value > 0.05 so we are assuming Normality, Independence and Ho-

moscedasticity.

Conclusion

The ANOVA p-value for Price is < 0.05, so at least one model has a significantly different mean log(Price). The Tukey HSD test shows which specific pairs differ. We expect $Boeing777$ and $AirbusA350$ (long-range) to have significantly higher prices than $Boeing737$ and $AirbusA320$ (medium-range).

2.3 Apply a two-way ANOVA including Sales Region to the model. Interpret your findings.

We applied ANOVA and the results are reported in table 2.1. There is no significant interaction since the p-value is much higher than 0.05, so we can conclude that the relationship between the aircraft Model and its Price does not change based on the Sales Region. Also, while there might be a tiny hint of a regional difference, statistically speaking, Region does not significantly affect Price.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Model	3	6032	2010.7	28038.546	< 2e-16 ***
SalesRegion	5	1	0.2	2.142	0.0575 .
Model:SalesRegion	15	1	0.1	0.830	0.6445
Residuals	8263	593	0.1		

Table 2.1: ANOVA for Price and Region

2.4 Convert the variable Production Year to a categorical variable with two levels as "Older" and "Newer" and save it as a new variable named "year_cat" in the data frame.

To compute the categorization of Production Year we first calculated the median and then applied it to a new variable: *year_cat*.

```
median = median(air_data$ProductionYear)
air_data$year_cat <- as.factor(
    ifelse(air_data$ProductionYear < median, "Older", "Newer")
)
```

2.5 Analyze the effect of Model and year ("year_cat") together on the price. Analyze whether the interaction of two term is significant. Interpret your findings.

Figure 2.5 plot aligns with previous analysis: there are two cheaper plane models and two expensive ones, and now we observe that there are older and newer models also which are a bit more expensive (shifted to the right) in each case. The ANOVA interaction plot shown in figure 2.6 is just one more analysis that make us sure about what has been deducted before.

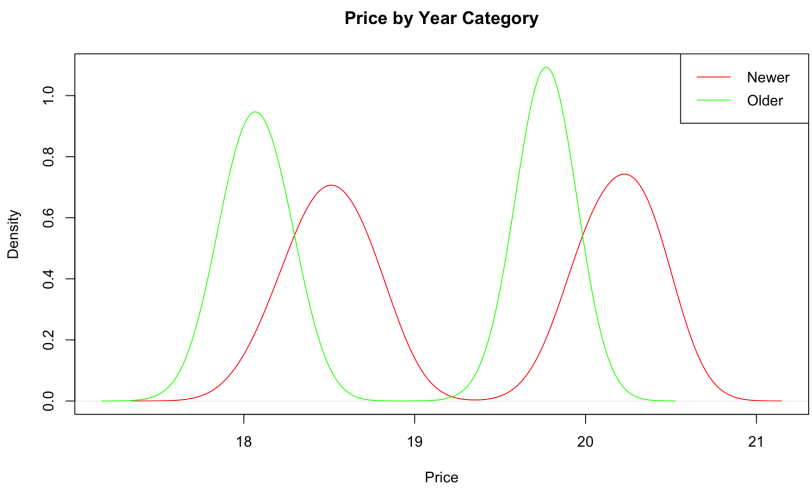

Figure 2.5: Plot by year_cat

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Model	3	6032	2010.7	78235.015	< 2e-16 ***
year_cat	1	381	381.4	14838.647	< 2e-16 ***
Model:year_cat	3	0	0.0	0.984	0.399
Residuals	8279	213	0.0		

Table 2.2: ANOVA for Price by Model and Year

Conclusion

The two-way ANOVA with interaction tests three effects:

Based on the two-way ANOVA results, both the airplane Model and the categorized production year (`year_cat`) have a significant independent effect on the Price. However, the interaction between the Model and the production year is not statistically significant, indicating that the price difference between newer and older airplanes remains consistent regardless of the specific airplane model.

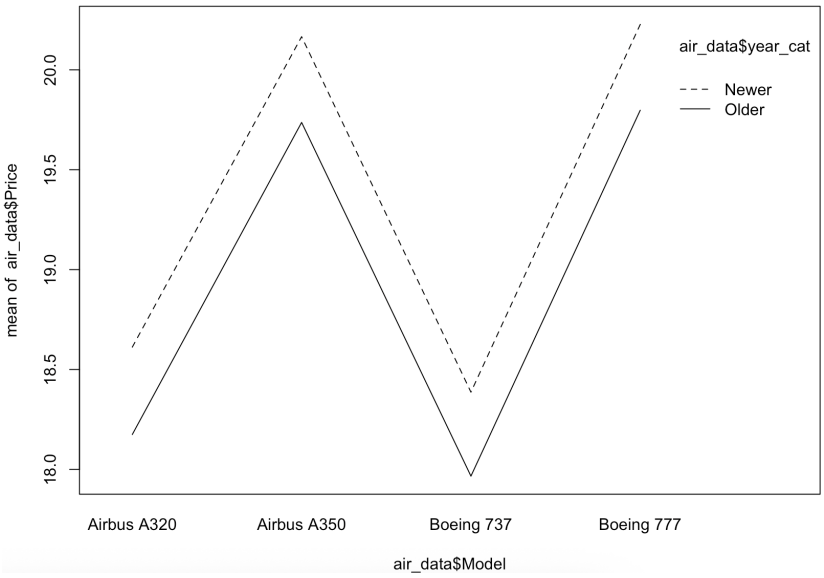

Figure 2.6: ANOVA Interaction Plot

Post-hoc tests

All three post-hoc tests (Tukey, Bonferroni, and LSD) confirm significant price differences between the "cheaper" models ($A320$, $B737$) and the "expensive" models ($A350$, $B777$). However, there is likely no significant price difference *between* the two cheaper models themselves, or *between* the two expensive models. The results are consistent across the different penalty methods.

Conclusion

Appendix A

Appendix Chapter

Use of AI

- Gemini (Pro): sentence enhancing, crosschecking personal assumptions